

Independent Application Answer Sheet

Name: _____ Date: _____

Directions

Use the lesson, worked examples, and the linked hydrogen-atom simulation. Record all answers on this sheet. Show substitutions, units, and reasoning. Submit this completed PDF with **U2 L7 Notebook Evidence Submission**.

Part A — Electromagnetic relationships

1. Put these regions in order from **longest wavelength to shortest wavelength**: ultraviolet, radio, visible, X-ray, infrared.

1	2	3	4	5

2. A blue-green photon has a wavelength of 486 nm. Convert 486 nm to meters, then calculate frequency using $c = \text{wavelength} \times \text{frequency}$ and $c = 3.00 \times 10^8 \text{ m/s}$.

Conversion to meters	Substitution	Frequency with unit

3. Use $E = h \times \text{frequency}$, where $h = 6.626 \times 10^{-34} \text{ J s}$, to calculate the energy of the photon from Question 2.

Substitution	Energy with unit

Part B - Atomic spectrum evidence

Open the linked PhET Models of the Hydrogen Atom simulation from the Moodle lesson. Use the same model and spectrum settings specified there.

Model/settings used	Two spectrum observations	What the evidence shows about atomic energy

4. Compare the line spectrum with a continuous rainbow. Explain why a line spectrum supports the idea that electron energies are quantized.

Response

5. Application: Choose astronomy, chemical identification, medical imaging, or communications. Explain how a specific region or spectrum measurement is used and cite one piece of scientific evidence from this lesson.

Application	Region or measurement	Evidence-based explanation

Final check: Every calculation includes units; every explanation connects evidence to wavelength, frequency, energy, or quantized atomic levels.